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POSTER

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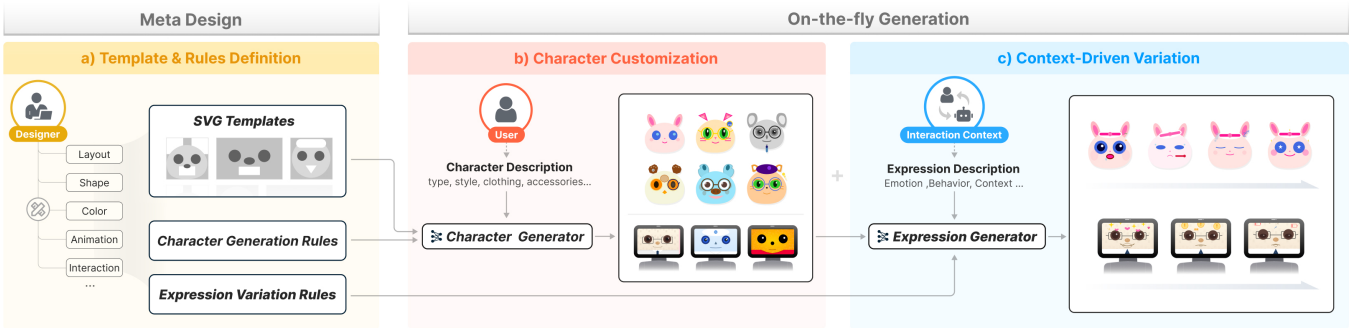

Figure 1: Conceptual framework of our approach. Meta design constrains the facial expression generation space; users participate explicitly or implicitly in character customization; the system adapts expressions based on interaction context.

Abstract

Facial expressions are a key component of intelligent agent interfaces, supporting social presence, character shaping, and context-sensitive interaction. This paper introduces a conceptual framework for generating personalized and adaptive facial expressions with large language models (LLMs). The framework includes three stages: meta-design, where designers define expression templates and transformation rules; character customization, where agents are shaped based on user input or role descriptions; and context-driven variation, where expressions dynamically respond to interaction context. We illustrate the framework through two cases: a virtual pet application that generates character-based emotional expressions from user profiles, and a service robot interface that adapts expressions in real time based on dialogue. This work-in-progress explores how LLMs can enable personalized and flexible facial expression interfaces for intelligent agents. Future work will focus on advancing the technical implementation and conducting user-centered evaluations.

CCS Concepts

• **Human-centered computing → Interaction techniques**.

Keywords

LLMs, Robot Face, Facial Expression, Human-Robot Interaction, Expressive Agents

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1 Introduction

Facial expression is a key component of social intelligence in intelligent agents, including both social robots and conversational agents.

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It plays a vital role in fostering user engagement and shaping the perceived personality and capabilities of the agent[2]. Among various approaches, rendered facial expressions displayed on screens offer high flexibility for defining the agent's character and social cues[4]. Dynamic adaptation to interaction context is crucial for enhancing social presence and perceived intelligence. Context-sensitive expressions that evolve with interaction improve expressivity and responsiveness[10–12]. In addition, customizable expression interfaces that reflect individual user preferences, instead of relying on pre-defined templates, can further enhance engagement and the agent's perceived social intelligence[5, 9]. With the rise of LLMs, these models are increasingly integrated into intelligent agents to support expressive behaviors that are more flexible, personalized, and contextually aligned[3, 8, 12, 13]. Among these, the Xpress system specifically demonstrates the use of LLMs for generating facial expressions in response to emotional and contextual cues[1].

In this paper, as illustrated in Figure 1, we propose a conceptual generative facial expression framework designed to support personalized human-agent interaction. The framework consists of three core stages. First, in the meta-design phase, designers constrain the space of facial expression generation by specifying rules and templates, which are subsequently embedded into the prompts used by the LLM-based generator. Second, during character customization, users participate either directly or indirectly in shaping the agent's character. Based on the character description, the system generates a personalized base facial expression. Third, during interaction, the facial interface adapts to the interaction context by varying its expressions according to the transformation rules defined in the meta-design phase. To demonstrate the framework, we describe two illustrative cases in this paper.

2 Case 1: Virtual Pet

Our first case demonstrates a virtual pet app that creates personalized agent characters based on user profiles. As shown in Figure 2, this app generates a customized virtual companion by taking user-provided information such as age, gender, occupation and MBTI type, and mapping it to a unique character design. The app also adapts the character's facial expressions according to changes in pet status, including hunger, fatigue, excitement, and other emotional states. To enable this personalization and adaptability, the system first processes the user profile using the Claude Opus 4 model to generate a textual character description. This description is then combined with an SVG template and predefined generation rules to produce the base character design. Building on this base, the system generates animated facial expressions that reflect rich emotional context across different interaction scenarios.

3 Case 2: Service Robot

Our second case demonstrates the use of our method on a service robot, as shown in Figure 3. The robot's facial interface can be customized based on its service role in a given scenario or shaped according to user intent, enabling flexible character design. During interaction, the system analyzes the dialogue in real time and adjusts the robot's facial expressions based on contextual cues. Technically, this case shares a similar implementation with the first one,

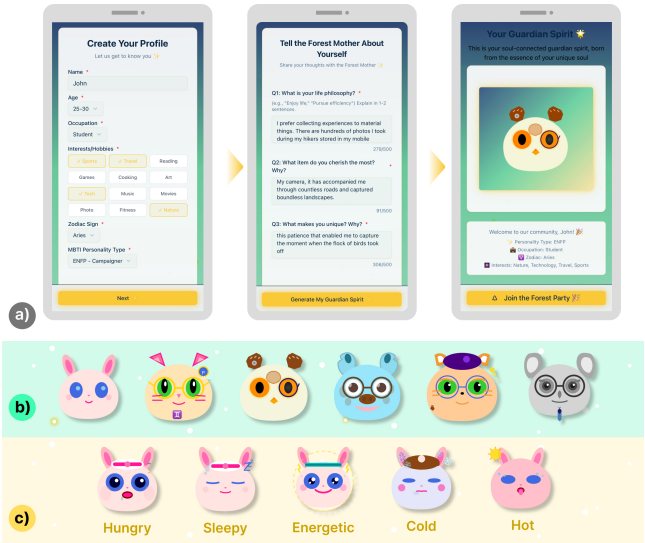

Figure 2: (a) Character customization via user input; (b) different base characters for different users; (c) facial expressions showing pet states.

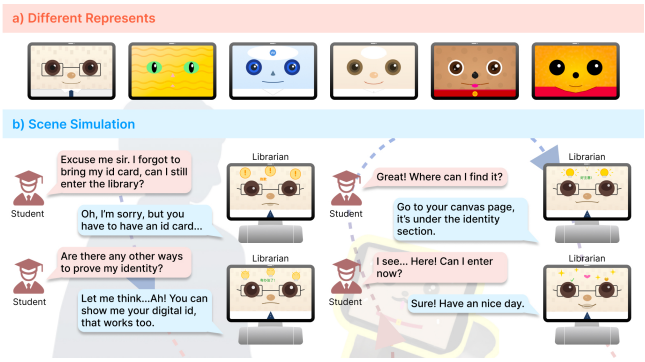

Figure 3: (a) Facial expression interfaces customized based on character-shaping needs; (b) expressions generated in real time according to dialogue context.

but differs in that users directly provide role descriptions, and the system uses live dialogue text as prompt input to drive expression generation.

4 Conclusion and Future Work

We presented a conceptual framework for generating personalized and context-aware facial expression interfaces for intelligent agents, demonstrated through a virtual pet app and a service robot interface. The framework includes meta-design, user-driven character customization, and context-driven expression variation during interaction. Future work will involve user studies in real-world settings to evaluate usability and interaction outcomes. In addition, we will explore how facial expression styles and levels of realism affect user perception and engagement, as discussed in recent work on embodied agents like Dittos [6, 7].

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